Power BI Assignment 2
DAX & Data Visualization

E-commerce sales analysis
A walkthrough of the model, DAX, report pages, and visual setup

Prepared from the supplied PBIX, CSV source files, and assignment brief
Submission edition | 9 report pages inspected

Project overview and result

The report analyzes Indian e-commerce orders across April 2018 to March 2019 using order headers, detail lines, and monthly category targets.

The PBIX contains nine named report pages. Eight pages hold one primary visual; the KPI page holds three card visuals.

Page bindings cover target comparison, margin, monthly sales, profit and quantity, cards, a matrix, geography, treemap, and state order counts.

CSV totals show ₹431,502 in Amount against ₹435,900 in targets, a ₹4,398 gap. Electronics exceeds target; Clothing has the largest shortfall.

₹431,502

SOURCE CSV AMOUNT TOTAL

₹435,900

SOURCE CSV TARGET TOTAL

-₹4,398 | 99.0%

Amounts use the Amount field, matching the scale of the monthly category targets.

Evidence and formula confidence

Verified from the PBIX report definition: nine page names, visual types, page size, positions, and fields assigned to visual roles.

Verified from the CSVs: source columns, row counts, date range, category totals, target totals, and sample rows.

The assignment brief names three calculated columns and three required measures. DAX below follows the stated logic.

The model is stored in an XPress9-compressed DataModel, and Power BI Desktop is unavailable here. Exact formula bodies and relationship settings could not be read verbatim.

Visuals in this deck are editable recreations from the PBIX bindings and CSV data, not screenshots captured from the Power BI window.

Formula reconstructions and visual recreations are labeled so the evidence stays clear.

Source files and model tables

Table	Rows	Fields / role
List of Orders	560 raw; 500 populated	Order ID, date, city, state, customer
Order Details	1,500	Amount, Profit, Quantity, Category, Sub-Category
Sales target	36	Month, Category, Target
Category Master	Model table	Category axis shared by actuals and targets
Calendar	Model table	Date and Month Year for monthly analysis
Category Count	Model object	Present in diagram; formula role not exposed
Order ID Count	Model object	Present in diagram; formula role not exposed

Likely join keys are Order ID, Category, and Month/Date. The report JSON exposes field use, while the compressed model hides relationship direction and cardinality.

Data quality and baseline checks

List of Orders has 560 CSV records, including 60 fully blank rows. Removing blanks leaves 500 unique populated Order IDs.

Order Details has 1,500 detail lines with those 500 order IDs. Sales target has 36 rows: 12 months by 3 categories.

Order dates span 1 April 2018 through 31 March 2019. The populated order headers have no duplicate keys.

Source totals: Amount ₹431,502; Profit ₹23,955; Quantity 5,615; Target ₹435,900.

Exclude blank order rows before order counts. A distinct count that includes blanks can add an extra value.

Main cleanup item: 60 entirely blank rows in List of Orders.

Table view: inspect loaded rows and columns

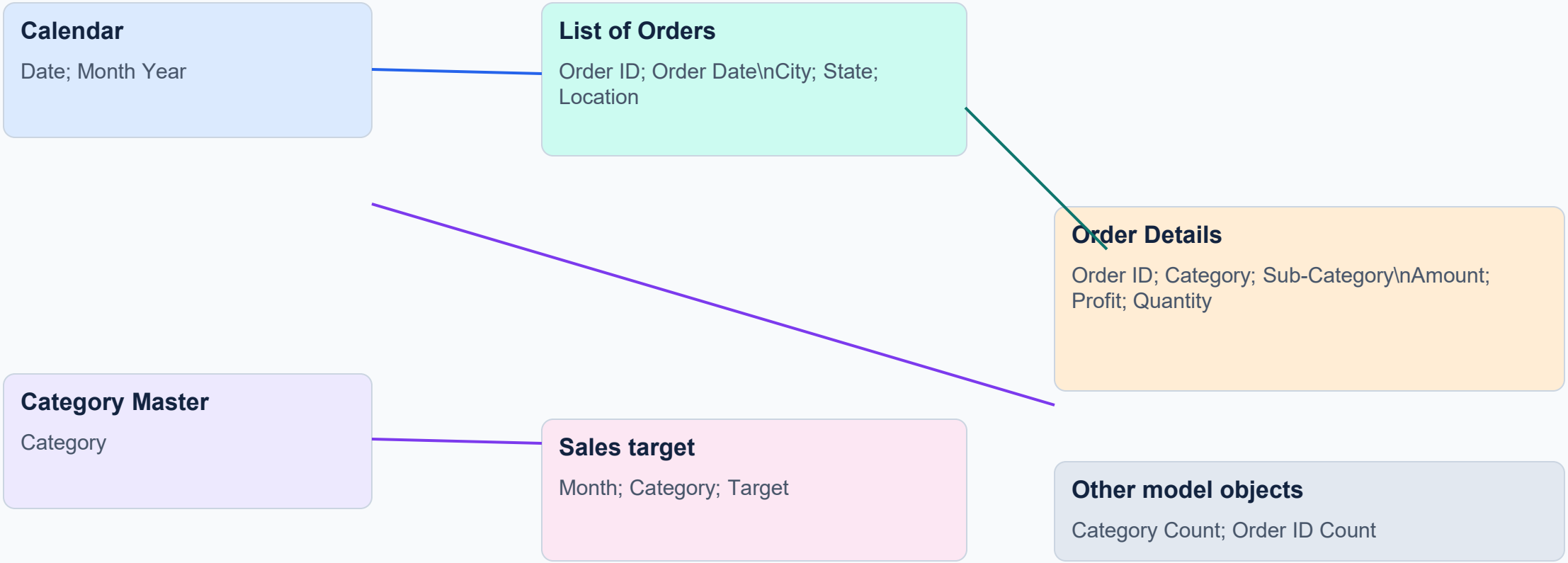
Order ID	Amount	Profit	Quantity	Category	Sub-Category
B-25601	1,275	-1,148	7	Furniture	Bookcases
B-25601	66	-12	5	Clothing	Stole
B-25601	8	-2	3	Clothing	Hankerchief
B-25602	80	-56	4	Clothing	Trousers

Select a table and inspect its columns, data types, blanks, and row values.

Create a calculated column from Table tools or Modeling. The new stored field appears in the grid.

Sort and filter rows to validate categories, dates, numeric ranges, and keys before charting.

Model view: table roles and join paths



Shown key paths are inferred from shared fields and report bindings; exact cardinality, filter direction, and active state were not extracted.

Calculated tables: Calendar and Category Master

CALENDAR | RECONSTRUCTED

```
Calendar =
CALENDAR(
    MIN('List of Orders'[Order Date]),
    MAX('List of Orders'[Order Date])
)
```

CATEGORY MASTER | RECONSTRUCTED

```
Category Master =
DISTINCT(
    'Order Details'[Category]
)
```

Calendar supplies the Date and Month Year fields used by the matrix and supports time calculations.

Category Master supplies the shared category axis used for actual sales compared with targets.

The model also lists Category Count and Order ID Count. Their formulas and roles are not visible in readable report metadata.

Both table expressions below are reconstructions from the observed fields.

Calculated columns: assignment and report fields

Field	Table	Purpose
Category Type	Order Details	Join Category and Sub-Category into one label
Revenue per Order	Order Details	Amount multiplied by Quantity
Sales Category	Order Details	Above / below average Amount classification
Month Year	List of Orders	Monthly trend category
Month Year	Calendar	Matrix month column
Location	List of Orders	Map location field with City and State

Formula bodies for report-bound fields are reconstructed and should be compared with the original model.

Calculated column 1: Category Type

RECONSTRUCTED DAX FROM THE ASSIGNMENT LOGIC

```
Category Type =
'Order Details'[Category] & " - " &
'Order Details'[Sub-Category]
```

Evaluates once per Order Details row and stores a combined product label, for example Clothing - Stole.

Use the field in table rows, slicers, filters, or visual categories when one label should identify both category levels.

The report visuals bind Sub-Category directly, so Category Type is an extra classification field.

Verify the separator against the formula bar if you need a verbatim expression.

Calculated column 2: Revenue per Order

RECONSTRUCTED DAX FROM THE ASSIGNMENT LOGIC

```
Revenue per Order =
'Order Details'[Amount] *
'Order Details'[Quantity]
```

The formula multiplies Amount by Quantity on each detail row.

Example from the first source row: 1,275 × 7 = 8,925.

A calculated column is stored at refresh and can be grouped or filtered like other numeric fields.

The target visuals align with SUM(Amount). Keep this required coursework field separate from the Amount-based sales total.

The brief explicitly requests Amount × Quantity; confirm Amount's business meaning before using this as report sales.

Calculated column 3: Sales Category

RECONSTRUCTED DAX FROM THE ASSIGNMENT LOGIC

```
Sales Category =
VAR AverageAmount =
    CALCULATE(
        AVERAGE('Order Details'[Amount]),
        REMOVEFILTERS('Order Details')
    )
RETURN
    IF('Order Details'[Amount] >= AverageAmount,
       "Above Average", "Below Average")
```

AverageAmount stores the overall average of Amount while REMOVEFILTERS clears the current row context.

IF compares each line's Amount to that threshold and assigns a text label.

The label is calculated at refresh. Slicers do not recalculate the stored category.

Reconstruction uses >=, so an Amount equal to average is Above Average.

Calculated columns and measures

CALCULATED COLUMN

Computes and stores a value per row.

Useful for Category Type, Sales Category, month labels, and Location.

Can be used in axes, slicers, rows, columns, and filters.

Recalculates during data refresh.

MEASURE

Calculates a result in the current filter context.

Useful for sales, targets, profit, quantity, and order counts.

The same measure changes by month, category, state, or sub-category.

Recalculates when report filters and selections change.

Columns describe rows. Measures summarize the rows selected by a visual.

Required measure 1: Order Count

RECONSTRUCTED DAX FROM THE ASSIGNMENT LOGIC

```
Order Count =
CALCULATE(
    DISTINCTCOUNT('Order Details'[Order ID]),
    'Order Details'[Order ID] <> BLANK()
)
```

Order Details has 1,500 line rows but 500 populated Order IDs.

DISTINCTCOUNT after excluding blanks returns one count per order.

COUNTROWS(Order Details) returns 1,500 and should be labeled Order Lines.

For the state funnel, the same distinct nonblank count should use List of Orders[Order ID].

Expected source count after blank removal: 500 populated orders.

Required measure 2: Average Profit in Delhi

RECONSTRUCTED DAX FROM THE ASSIGNMENT LOGIC

```
Average Profit in Delhi =
CALCULATE(
    AVERAGE('Order Details'[Profit]),
    'List of Orders'[City] = "Delhi"
)
```

AVERAGE summarizes profit at order-detail line level.

CALCULATE applies the City = Delhi filter before the average is computed.

The order header to detail relationship must carry the city filter into Order Details.

If the requirement means average profit per order, aggregate by Order ID before averaging.

Reconstructed from the assignment logic; confirm the relationship in Model view.

Required measure 3: Year-to-date Sales

RECONSTRUCTED DAX FROM THE ASSIGNMENT LOGIC

```
YTD Sales =
TOTALYTD(
    [Total Sales],
    'Calendar'[Date]
)
```

TOTALYTD accumulates the base sales measure from the start of the calendar year through the current date context.

Calendar must be continuous and related to order dates for the time calculation.

Standard YTD resets in January. A running total from the first dataset date does not reset.

The supplied data spans April 2018 to March 2019, so standard YTD restarts in January 2019.

YTD and all-history running total have different meanings.

Report measures: DAX patterns used by visuals

SALES, TARGET, PROFIT, AND QUANTITY

```
Actual Sales = SUM('Order Details'[Amount])
Sales Target = SUM('Sales target'[Target])
Total Profit = SUM('Order Details'[Profit])
Total Quantity = SUM('Order Details'[Quantity])
Max Profit Margin = MAXX('Order Details', DIVIDE([Profit], [Amount]))
```

Actual Sales / Sales Target appear in the clustered column chart and cards.

Profit and Quantity feed the scatter chart. Max Profit Margin feeds the donut.

Total Actual Sales and Total Sales Target are report measure names with equivalent total patterns.

Visualizations pane: field roles for pages 1 to 5

Visual	Role	PBIX field binding
Clustered column	Category / Y values	Category Master[Category] / Actual Sales; Sales Target
Donut	Category / values	Order Details[Sub-Category] / Max Profit Margin
Line	Category, series, Y	List of Orders[Month Year] / same / Total Sales
Scatter	X / Y / details	Total Quantity / Total Profit / Sub-Category
Cards	Data / rows	Total Actual Sales; Total Sales Target; Minimum Target by Category

These role mappings come from the PBIX query definitions, not from a screen capture.

Visualizations pane: field roles for pages 6 to 9

Visual	Role	PBIX field binding
Matrix	Rows / columns	Category Master[Category] / Calendar[Month Year]
Matrix	Values	Total Actual Sales; Total Sales Target; Sales Variance; Target Achievement %
Map	Location / size	List of Orders[Location] / Total Sales
Map	Geography / tooltip	City; State; Total Sales
Treemap	Group / values	Order Details[Sub-Category] / Total Sales
Funnel	Category / values	List of Orders[State] / Total Order Count

Use a shared category and date dimension for consistent grouping.

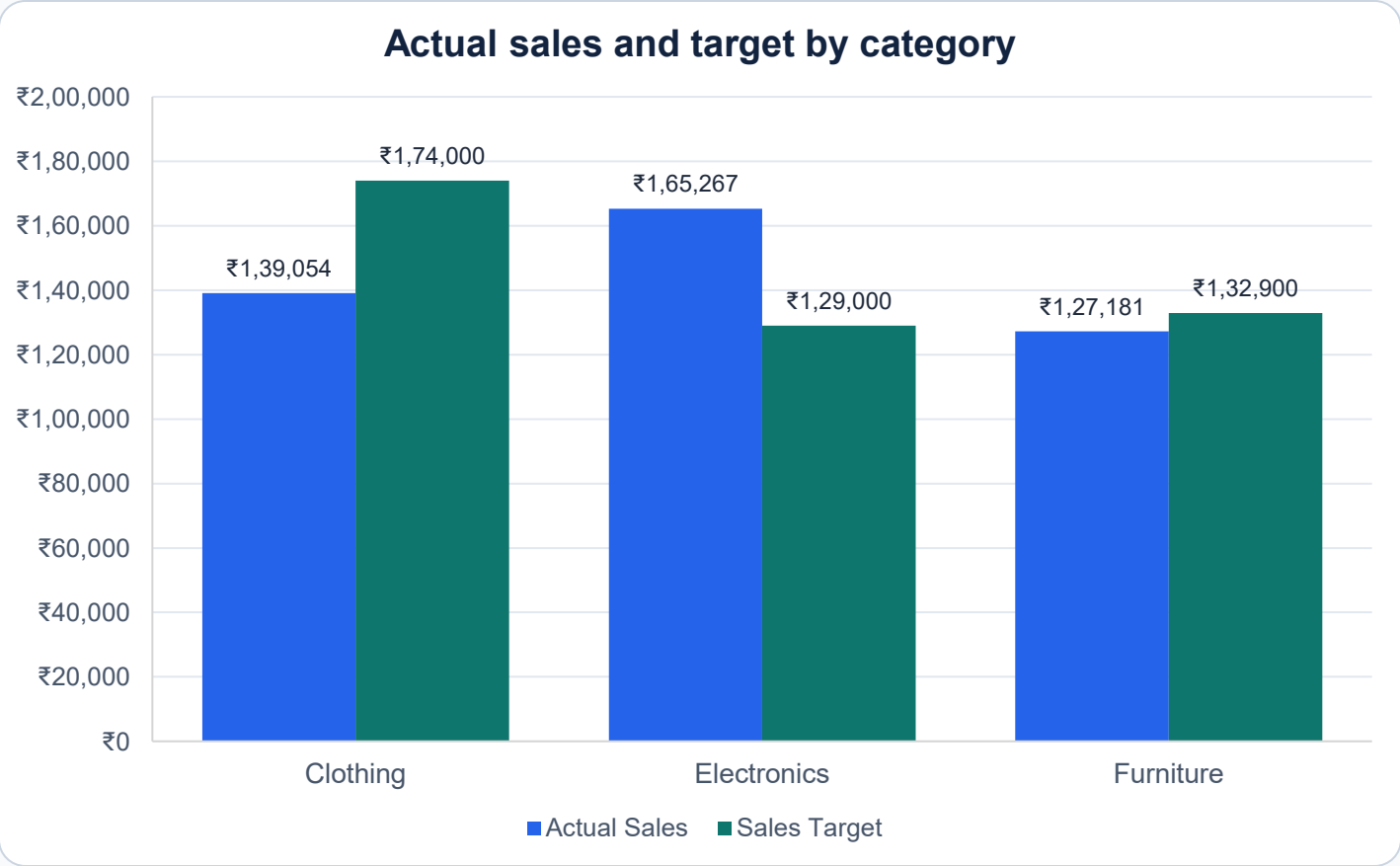
Use measures for numeric values, not raw text columns.

Check geographic categories and blank keys before using Map.

Report view: nine pages in the PBIX

Page	Page name	Primary visual
1	Sales Target Achievement by Category	Clustered column
2	Max Profit Margin by Sub-category	Donut
3	Monthly Sales Trend	Line
4	Comparison of Profit and Quantity by Sub-Category	Scatter
5	Comparison of Total Sales Amount and Target	Three cards
6	Sales Performance Matrix	Matrix
7	Geographic Sales Analysis	Map
8	Sales Distribution by Sub-Category	Treemap
9	Order Count Analysis by State	Funnel

Visualization 1: Sales Target Achievement by Category



Fields: Category Master[Category], Actual Sales, and Sales Target.

Clothing: ₹139,054 actual versus ₹174,000 target, 20.1% below target.

Electronics: ₹165,267 actual versus ₹129,000 target, 28.1% above.

Furniture: ₹127,181 actual versus ₹132,900 target, 4.3% below.

Editable recreation from report bindings and CSV totals, not a Power BI UI screenshot.

Visualization 2: Max Profit Margin by Sub-category

Maximum row-level margin by sub-category



Fields: Sub-Category as category; Max Profit Margin as value.

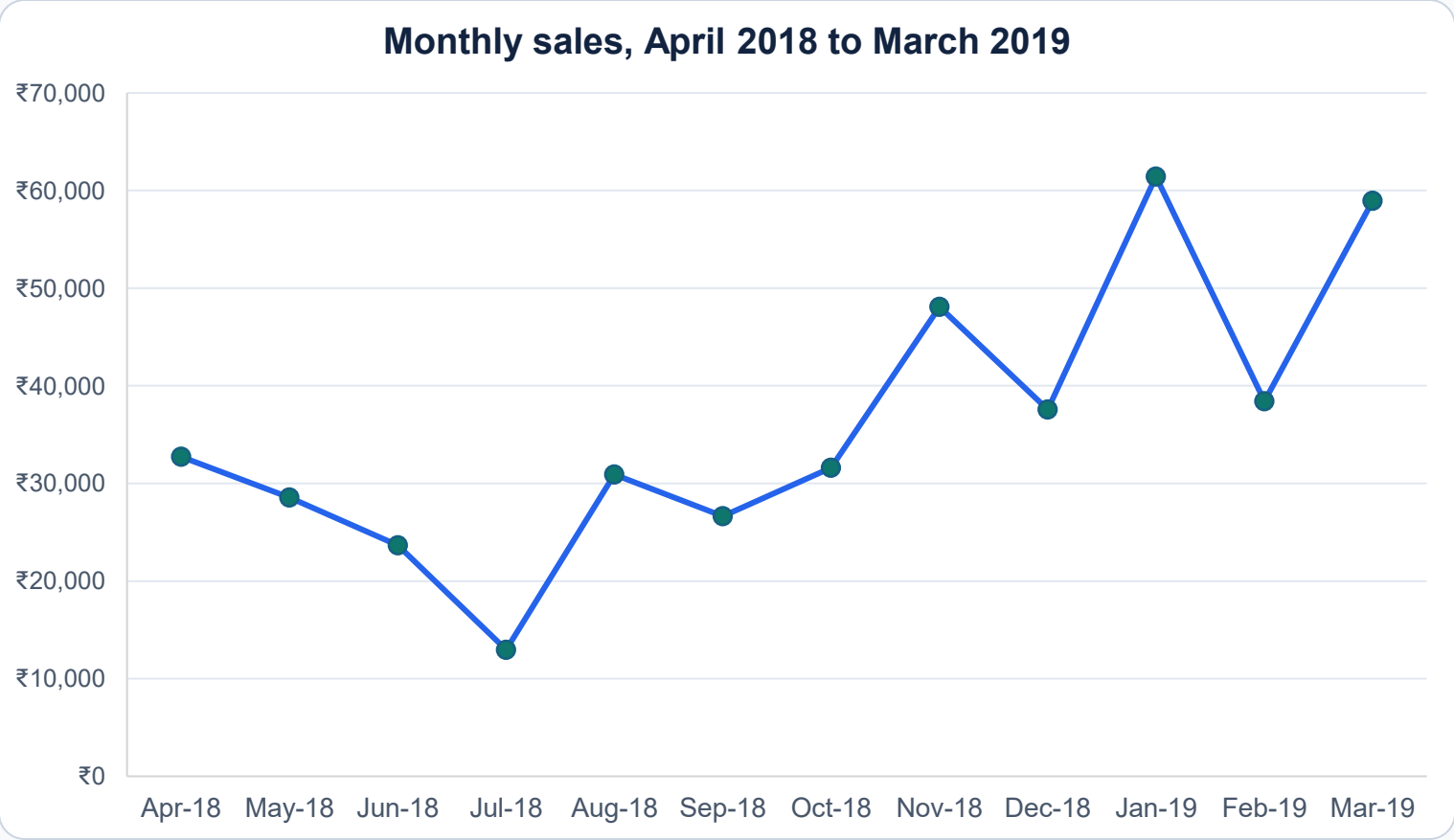
Preview includes all 17 sub-categories from the source table.

Highest reconstructed maximum ratios: Leggings 50.85%, Stole 50.62%, Trousers 49.96%.

A maximum ratio is an extreme row value, not average profitability.

Margin preview uses MAX(Profit / Amount) per sub-category; verify the original measure.

Visualization 3: Monthly Sales Trend



Fields: List of Orders[Month Year] on the month axis and Total Sales on Y.

January 2019 peaks at ₹61,439; March reaches ₹58,937.

July 2018 is the minimum at ₹12,966.

Sort Month Year by date or month number to keep April-to-March order.

Monthly totals join detail lines to populated order headers by Order ID.

Visualization 4: Profit and Quantity by Sub-category



- Fields: Total Quantity on X, Total Profit on Y, Sub-Category as point detail.
- Tables totals -₹4,011; Electronic Games totals -₹1,236.
- Saree and Hankerchief have high quantities but smaller total profits.
- Label selected outliers rather than every point.

All 1,500 order detail lines are aggregated by sub-category.

Visualization 5: Total Sales and Target cards

₹431,502

TOTAL ACTUAL SALES

₹435,900

TOTAL SALES TARGET

Gap -₹4,398 | Achievement 99.0%

Category	Minimum monthly target
Clothing	₹12,000
Electronics	₹9,000
Furniture	₹10,400

PBIX KPI page binds Total Actual Sales, Total Sales Target, and Minimum Target by Category.

Visualization 6: Sales Performance Matrix

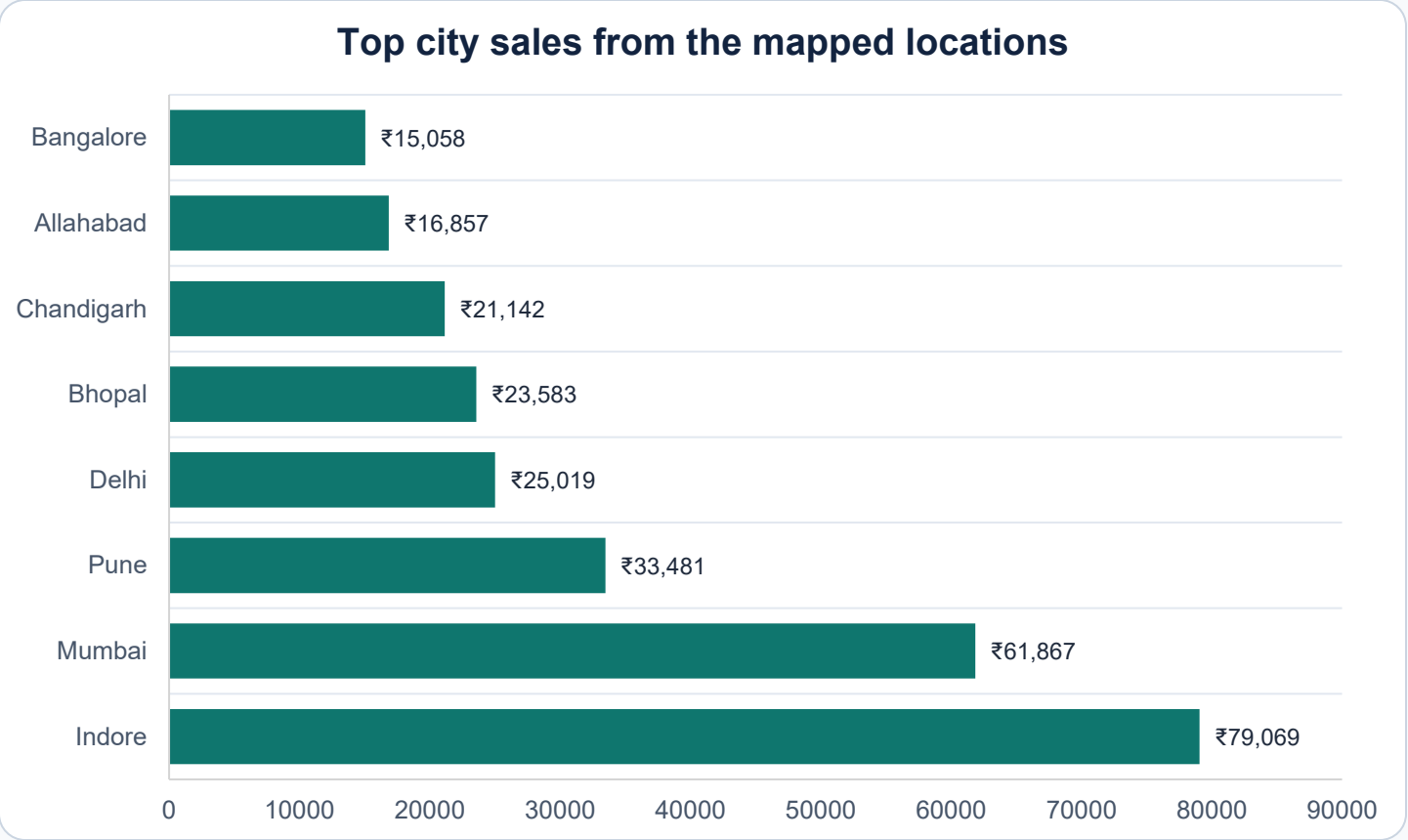
Category	Month	Actual	Target	Variance	Achievement
Clothing	Apr-18	₹13,478	₹12,000	₹1,478	112.3%
Clothing	May-18	₹9,518	₹12,000	-₹2,482	79.3%
Electronics	Apr-18	₹11,127	₹9,000	₹2,127	123.6%
Electronics	May-18	₹12,807	₹9,000	₹3,807	142.3%
Furniture	Apr-18	₹8,121	₹10,400	-₹2,279	78.1%
Furniture	May-18	₹6,220	₹10,500	-₹4,280	59.2%

Rows: Category Master[Category]. Columns: Calendar[Month Year].

Values: actual sales, target, variance, and target achievement percentage.

Each cell reflects the filter context at one category and one month.

Visualization 7: Geographic Sales Analysis



PBIX bindings include City, State, Location, and Total Sales.

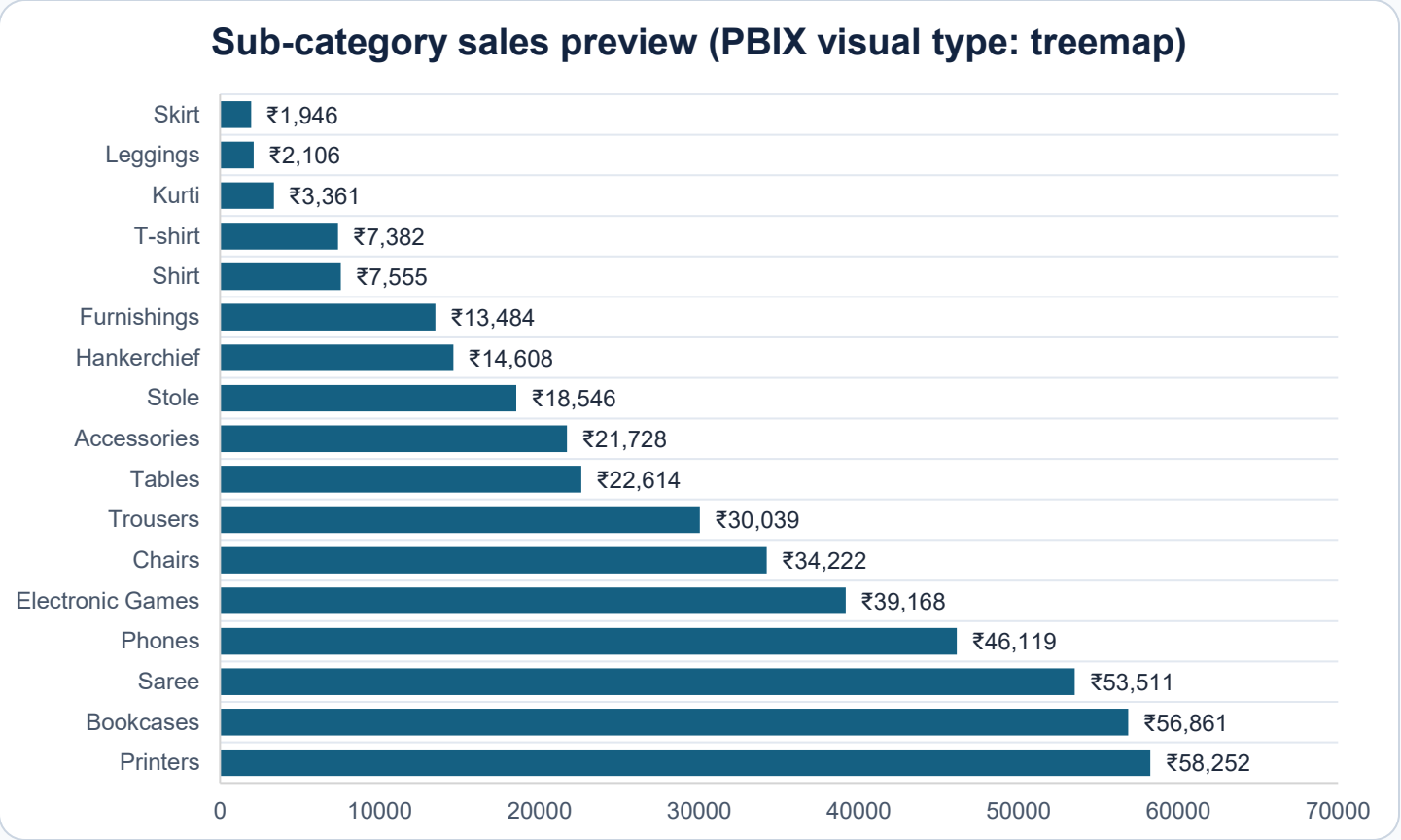
Indore leads at ₹79,069; Mumbai follows at ₹61,867.

The report's map screenshot is unavailable here; this is a top-city value preview.

Set geographic data categories and confirm city/state resolution in Power BI.

Report field bindings are extracted; the chart preview is not a map or UI screenshot.

Visualization 8: Sales Distribution by Sub-category



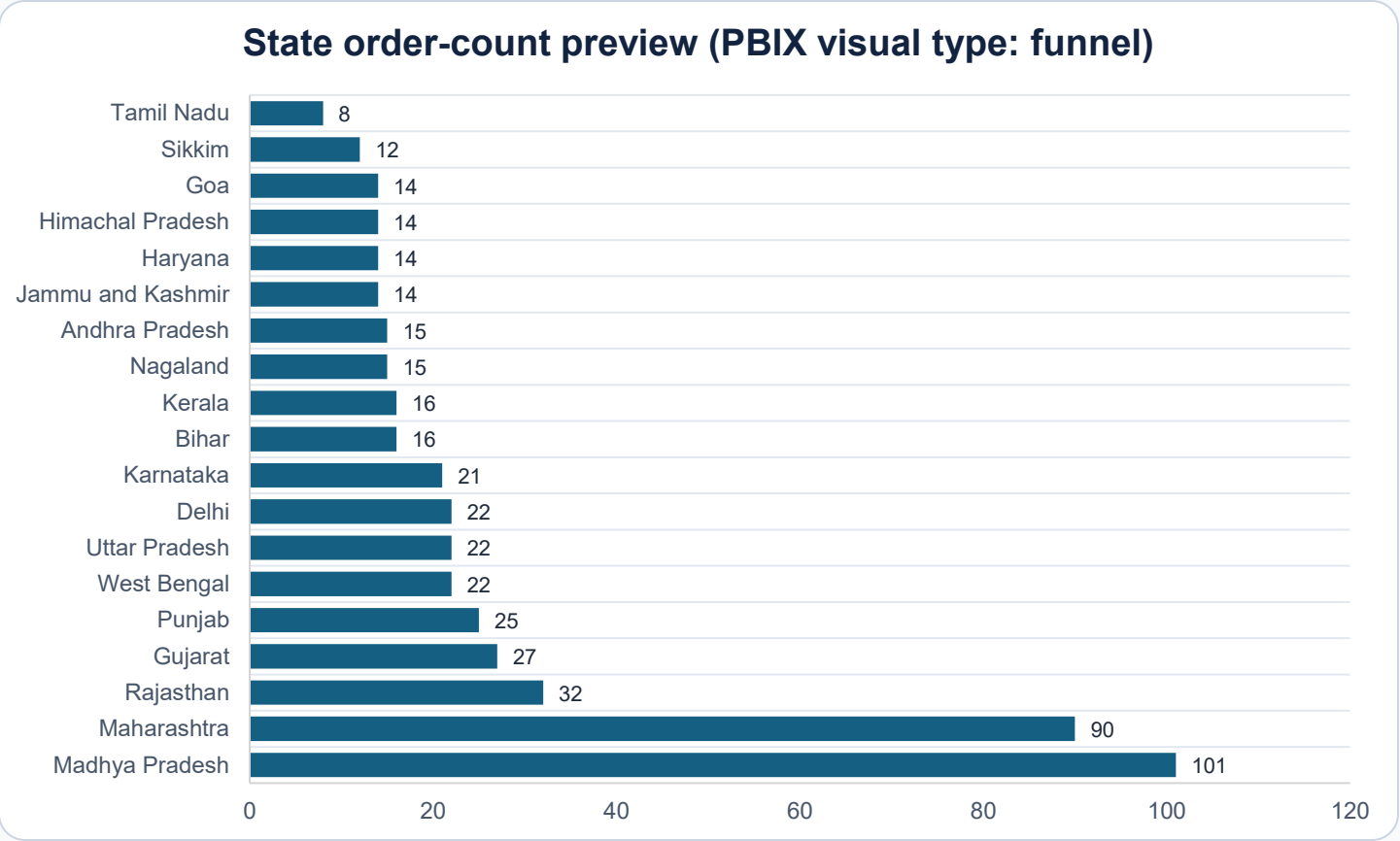
PBIX fields: Sub-Category in Group and Total Sales in Values.

Printers lead at ₹58,252, then Bookcases at ₹56,861 and Saree at ₹53,511.

Tile size represents sales; color should not imply profit unless it encodes profit.

PBIX visual type: treemap. This editable bar preview shows the same sub-category sales grouping.

Visualization 9: Order Count Analysis by State



PBIX fields: State on Category and Total Order Count on Y.

Madhya Pradesh has 101 populated orders; Maharashtra has 90.

The state counts sum to 500 after excluding blank order rows.

A sorted bar chart can be easier to compare across independent states.

PBIX visual type: funnel. This editable bar preview shows order counts by state for direct comparison.

Step-by-step: Table view, Model view, and Report view

View	Steps	Check
Table view	Choose table; inspect rows and types; filter blanks; create DAX columns/measures.	500 populated IDs; 1,500 detail rows; remove 60 blank order rows.
Model view	Connect matching keys; inspect cardinality and direction; mark Calendar as a date table.	Order ID, category, and month/date paths support report filtering.
Report view	Select page; choose visual; assign fields; format titles, axes, labels, and colors.	Nine pages match the assignment's requested chart topics.

Complete data checks and relationships before building visuals.

Test category, month, and state selections to confirm intended filter behavior.

Save after checking totals, sorting, refresh status, and number formats.

Step-by-step: build and format each visual

1. Choose a visual that matches the question: comparison, trend, distribution, relationship, geography, or summary.
2. In Build visual, assign category fields to axis/group roles and measures to values, X, Y, size, or tooltip roles.
3. Confirm aggregation: SUM for sales/profit/quantity; distinct nonblank count for orders; MIN for target floor; DIVIDE for achievement.
4. In Format visual, set a descriptive title, labels, axis titles, currency or percentage display, legend placement, and consistent colors.
5. Check month sorting, blank categories, map resolution, negative profit, and filtered results.
6. Use tooltips for detail and keep labels readable.

Power BI may show build and format controls in a pane or directly on the visual.

Project sequence, quality checks, and sources

1. Load the CSVs, confirm data types, and remove 60 blank order-header rows.
2. Confirm the populated Order ID key and relationship to the 1,500 detail lines.
3. Create Calendar and Category Master; verify category and date paths.
4. Add the three required columns and measures. Compare reconstructed expressions with the PBIX formula bar if available.
5. Build the nine report pages and field roles shown in this deck. Format currency, percentages, titles, legends, and month sorting.
6. Validate totals: Amount ₹431,502; Target ₹435,900; gap -₹4,398. Save the PBIX and use this presentation or PDF for submission.

Sources: supplied PBIX, three CSVs, assignment brief, and Microsoft Learn pages on DAX, calculated columns, date tables, and visual formatting.

This deck labels extracted facts separately from reconstructed formulas and visuals.